

ASSIGNMENT: IP ADDRESSING & SUBNETTING

Computer Networks & Infrastructure Analysis
Status: Pending Submission

Task 1: IP Address Classes & Real-World Examples

The IPv4 address space is historically divided into five distinct classes (A through E) based on the first few bits of the address. Each class serves a specific scale of network infrastructure, ranging from massive government networks to experimental multicast protocols.

Class	Leading Bits	Default Subnet Mask	Real-World Example / Use Case
Class A	0	255.0.0.0	Large enterprise or government core networks (e.g., General Electric / IBM internal WANs)
Class B	10	255.255.0.0	Medium-to-large university campus networks or multinational corporate offices
Class C	110	255.255.255.0	Small Office / Home Office (SOHO) routers and local residential Wi-Fi networks
Class D	1110	N/A (Multicast)	IP Multicast streaming (e.g., IPTV broadcasting or live corporate video town halls)
Class E	1111	N/A (Experimental)	R&D, experimental protocol testing, and future network architecture research

Task 2: IPv4 Address Calculations

Given the IP address 192.168.10.45 with a subnet mask of 255.255.255.0, perform the structural calculation for the network address, broadcast address, and valid host range.

- Working Steps & Calculations:
1. Binary Conversion:

- IP Address (192.168.10.45) : 11000000 . 10101000 . 00001010 . 00101101
- Subnet Mask (255.255.255.0): 11111111 . 11111111 . 11111111 . 00000000

2. Network Address Calculation (Bitwise AND operation between IP and Subnet Mask):

- 192.168.10. (Binary AND with 255.255.255) -> 192.168.10
- Host Octet (45 AND 0) -> 0
- Resulting Network Address: 192.168.10.0

3. Broadcast Address Calculation (Set all host bits to 1):

- Network Prefix: 192.168.10
- Host Octet set to binary 11111111 (Decimal 255)
- Resulting Broadcast Address: 192.168.10.255

4. Range of Valid Host Addresses (Network Address + 1 to Broadcast Address - 1):

- First Usable IP: 192.168.10.1
- Last Usable IP: 192.168.10.254

Task 3: Subnetting Plan for a Food Delivery Startup (Zomato Case Study)

A food delivery platform requires segregation of traffic into 4 independent subnets for security and operational control: Delivery Partners, Restaurants, Customers, and Admins. Using the base network 10.0.0.0/24, we subnet to accommodate these 4 groups.

Group Name	Subnet / CIDR	Subnet Mask	Usable IP Range	Total Hosts
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	62
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	62
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	62
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	62

Task 4: College WiFi Network Subnetting

A college WiFi network operates on the base range 172.16.0.0/16. The network administrator requires at least 20 subnets for different campus buildings. We must calculate the new subnet mask and available host capacity.

Calculation Steps & Constraints:

1. Determine Subnet Bits Required:

- Target Subnets = At least 20.
- Using formula $2^n \geq \text{Required Subnets}$, where n is the number of borrowed bits.
- $2^4 = 16$ (Insufficient, < 20)

- $2^{20} = 32$ (Sufficient, ≥ 20). Therefore, we must borrow 5 bits from the host portion.

2. Calculate New Subnet Mask:

- Original CIDR: /16 (Subnet mask: 255.255.0.0)
- New CIDR: /16 + 5 borrowed bits = /21
- Binary representation of third octet: 11111000 (Decimal 248)
- Final New Subnet Mask: 255.255.248.0

3. Calculate Available Hosts Per Subnet:

- Total bits in IPv4 address = 32
- Network + Subnet bits = 21 bits, leaving $32 - 21 = 11$ host bits.
- Total IP addresses per subnet = $2^{11} = 2048$
- Usable Host Addresses = $2048 - 2$ (subtracting Network ID and Broadcast Address) = 2,046 hosts per subnet.

Task 5: Troubleshooting an Invalid IP Address

A friend reports that their device configured with IP address 192.168.1.300 cannot connect to the local home network.

Error Identification & Explanation:

- The Error: In an IPv4 dotted-decimal address, each of the four octets represents an 8-bit binary number ranging from 0 to 255.
- The value '300' in the final octet exceeds the maximum decimal limit of 255 (since binary 11111011 = 251, and 11111111 = 255). A value of 300 requires 9 bits, which is mathematically invalid for an IPv4 octet.
- Solution / Valid Range: To successfully connect to the 192.168.1.0/24 network, the final octet must be changed to any integer between 1 and 254 (assuming 0 is the network ID and 255 is the broadcast address).
- Example of a Valid IP: 192.168.1.50